

1 (In)determinacy in woody plants: limits and opportunities for
2 timing growth in a changing climate

3 Frederik Baumgarten¹⁻³, Sally Aitken¹, Yann Vitasse³, Robert D Guy¹, EM Wolkovich¹

4 May 7, 2026

5 ¹ Department of Forest and Conservation Sciences, Faculty of Forestry, University of British Columbia,
6 2424 Main Mall Vancouver, BC Canada V6T 1Z4.

7 ² Department of Environmental Sciences-Botany, University of Basel, Schönbeinstrasse 6, Basel 4056,
8 Switzerland.

9 ³ Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Zürcherstrasse. 111, Bir-
10 mensdorf 8903, Switzerland

11 **Corresponding Author:** Frederik Baumgarten; frederik.baumgarten@unibas.ch;
12 Tel.: +41 61 207 23 37

13 **E-Mail-Addresses:** sally.aitken@ubc.ca; yann.vitasse@wsl.ch; rob.guy@ubc.ca; e.wolkovich@ubc.ca

14 **Type of article:** Perspective

15 **Short running title:** (In)determinate growth strategies of trees

16 **Keywords:** plant growth, tree phenology, shoot extension, indeterminate growers, carbon sequestra-
17 tion, growing season length, drought, genetic programming, phenotypic plasticity, forest productivity

18 **wordnumbers:** abstract: 200; main text: 4005; box1: 404; number of figures: 4; number of tables: 0

19 **number of references:** 96

20 **Authors contribution:** FB and EM initiated this project. FB took a lead in drafting, designed
21 figures and performed phylogenetic analyses. All authors contributed substantially to the writing of
22 the article.

23 **Data accessibility statement:** The data supporting the phylogeny are archived in the following
24 public repository. Link: <https://doi.org/10.5281/zenodo.18637216>.

25 **Conflict of interest statement:** The authors declare no conflict of interest

Abstract

When and how much plants grow under environmental constraints are fundamental questions in biology and increasingly important for predicting biomass production and carbon sequestration under climate change. While temperature and water availability directly regulate plant growth, the timing and rate of growth are also shaped by internal developmental programming, though this is rarely considered in predictions of tree growth responses to future climates.

Here, we revisit a concept central to this internal programming—(in)determinacy. Focusing on woody plants, we define it as the extent to which annual shoot growth is deployed from preformed organs versus produced *de novo* during the current season. We argue that this trait is best understood as a continuum and that it can help explain contrasting growth responses in a changing climate.

More determinate species concentrate growth within a narrow seasonal window, which may reduce exposure to late-season stress but also limit their ability to exploit longer growing seasons. More indeterminate species retain greater flexibility to extend or resume growth when conditions remain favorable, which may be advantageous under climate change, but this same flexibility may also increase exposure to frost, drought, and incomplete tissue maturation. Because primary shoot growth also shapes canopy development and is linked to other growth processes, variation in (in)determinacy could help explain broader differences in whole-plant performance, carbon gain, and species responses to climate change.

Introduction

Investing the right amount of resources in growth at the right time is of critical importance to the survival and fitness of any living organism. The topic of growth strategies and habits has a long history in science, spanning the fields of genetics, genomics, physiology and ecology across the animal and plant kingdoms (Stearns, 1998). Central to this discussion is the concept of determinacy, or more broadly (in)determinacy, which describes the extent to which growth is fixed or remains open over time. At its broadest, determinacy has been used to classify organisms according to whether growth ceases at maturity or continues throughout life; in this sense, plants are often described as indeterminate growers because they can continue adding biomass throughout their lives (Ejzmond *et al.*, 2010; Karkach, 2006). In plants, however, the concept has also been applied at finer spatial and temporal scales, from single cells and organs to whole shoots and individuals, and from a single season to an entire lifetime (McDaniel, 1992; Karkach, 2006).

In this Perspective, we focus on this narrower developmental meaning. In woody perennials, organs can develop from preformed tissue (e.g. primordia in overwintering buds), form *de novo*, or both, depending on species and life form. While these underlying developmental and meristematic constraints apply broadly across the plant kingdom, they play a particularly central role in the life cycle of woody perennials, especially trees growing in extra-tropical regions, which must balance growth, survival, and reproduction over long lifespans in highly seasonal environments. As we use it here, (in)determinacy in woody perennials refers to the ability to:

- a) preform organs in one year as a future investment, which is deployed rapidly in the following (or later) spring with no further primary shoot growth during that season (determinate strategy);
- b) maintain continuous or episodic meristem activity by forming new shoot tissue during the current growing season (indeterminate strategy).

In this Perspective, we revisit classical concepts of determinacy as they relate to shoot growth in woody perennials, place them in the context of climate change, and explore to what extent related developmental principles may also inform growth processes in other meristematic tissues. Although we focus on woody perennials, the concept of (in)determinacy is broadly relevant across vascular plants, including herbaceous species; trees provide a particularly informative model system because the ecological

76 and climatic consequences of growth timing are especially pronounced in long-lived woody plants. To
77 provide context, we first outline the fundamental limitations to plant growth in extra-tropical regions
78 by linking external environmental drivers with internal growth control mechanisms. In this context,
79 our aim is to provide a conceptual and methodological basis for analysing (in)determinacy as a trait
80 relevant to plant responses to climatic variability and change, to clarify how the concept can be defined
81 and measured, and to identify key research gaps and directions for future comparative and trait-based
82 analyses.

84 Seasonality of temperature, soil moisture and light

85 The further one travels from the equator towards the poles, the more strongly plants are confined to
86 a shrinking temporal window of opportunity for growth set primarily by low temperatures. Below c.
87 5 °C, metabolic activity slows to an extent where growth and development largely cease (Schenker
88 *et al.*, 2014; Rossi *et al.*, 2008; Körner, 2008). More importantly, sub-freezing temperatures can cause
89 severe damage to plant tissue if they occur at vulnerable developmental stages, e.g. during or after
90 leaf unfolding, prior to fruit maturation, or before cold acclimation in fall (Sakai & Larcher, 1987;
91 Baumgarten *et al.*, 2023). While annual plant species accommodate their entire life cycle within this
92 window, perennial (woody) species of extra-tropical regions are forced to partition their growth phase
93 seasonally, with periods of activity alternating with periods of dormancy (Delpierre *et al.*, 2016). This
94 is referred to as intermittent or rhythmic (as opposed to continuous) growth.

95
96 High temperatures can directly inhibit growth by exceeding species-specific physiological thresholds
97 (O’Sullivan *et al.*, 2017). In addition, increased evaporative demand and reduced precipitation can
98 lower soil moisture, leading to water stress that slows or stalls growth and, in extreme cases, causes
99 leaf scorching or premature senescence (Hsiao, 1973; Pugnaire *et al.*, 1999; Etzold *et al.*, 2021; Estiarte
100 & Peñuelas, 2015). Together, these temperature and soil moisture limitations act as environmental
101 filters, narrowing the period during which growth is possible (Figure 1). Notably, comparable (or even
102 stronger) seasonal constraints occur in seasonally dry forests as is typical in Mediterranean climates,
103 where temperatures may remain suitable for growth year-round but multi-month dry seasons impose
104 pronounced water limitation. Across species, leaf flush and renewed growth commonly track the onset
105 of rains and access to plant-available water (including deep soil moisture), making soil water availabil-

ity a primary driver of seasonal activity in these systems (Borchert, 1994; Ramos *et al.*, 2023; Alvarado *et al.*, 2025).

Light also influences plant growth through two key processes. First, light intensity provides the primary energy source that drives photosynthesis and, consequently, source activity. Second, light carries a wealth of information through its quality and the length of daily light and dark periods (photoperiod), which mediate many physiological processes, including bud set and dormancy transitions (Wang *et al.*, 2024). Both factors may contribute to the common observation that tree growth often peaks near the summer solstice in extra-tropical regions (Rossi *et al.*, 2006; Etzold *et al.*, 2021; Luo *et al.*, 2018).

Internal programming of plants

Given our relatively robust understanding of environmental influences on plant growth, it may seem straightforward to predict how plants will grow under current and future climatic conditions. Yet, this has proven challenging, particularly when predicting responses under scenarios with extended, climatically favorable growing seasons (Zohner *et al.*, 2021). Here we propose that accurate predictions require incorporating another key, but often overlooked, factor: internal growth control—the genetically encoded developmental program implemented through hormonal and gene-regulatory networks, that may prevent further growth *despite* favorable environmental conditions.

While plants have evolved numerous morphological adaptations to tolerate or avoid harmful environmental conditions, most temperate and boreal woody perennials cope with seasonal fluctuations in temperature and moisture by temporally escaping these conditions. They do this by aligning the timing of their life-history events (phenology) with a cycle of growth and dormancy that balances survival, reproduction, and growth over the long lifespan typical of many tree species (Delpierre *et al.*, 2016). Hence, the phenological sequence can impose internal switches in resource allocation—from vegetative growth to reproduction (flowering, fruit maturation) and storage (Stearns, 1989; Chapin *et al.*, 1990). These transitions act as additional internal filters, further narrowing the effective window during which vegetative growth can occur, regardless of environmental potential (Figure 1).

The concept of (in)determinate growth

In annual plants, growth ends with the production of flowers to form fruits and seeds: a signal in the apical meristem triggers a switch in resource investment from vegetative growth to building a reproductive structure with no point of return, signaling the end of the life-cycle (Poethig, 2003; Huijser & Schmid, 2011). A shoot axis that terminates in a flower or other modified structure and ceases meristematic activity is considered ‘determinate’ (Barthélémy & Caraglio, 2007). In woody perennial species, however, flowers are typically produced on lateral shoots or buds, preserving the main vegetative axis and allowing continued structural expansion. However, ‘determinate growth’ also refers to a temporal suspension of the meristem, which leads to a time lag between the formation of organs and their expansion (Kozłowski, 2012; Hallé *et al.*, 1978).

In their first year, all woody seedlings exhibit indeterminate growth. They begin growth from an embryonic shoot meristem and typically prioritize rapid leaf and canopy establishment, reflecting a juvenile, largely opportunistic phase of development (Borchert, 1976). In subsequent years, many temperate and boreal tree species preform their new shoot increments during the previous growing season, overwintering them as primordial shoots with leaves and internodes in hardened buds to be ‘ready-to-go’ when spring arrives. Once the leaves flush and the preformed internodes elongate, species differ in how long the shoot apical meristem remains competent to continue organ production (Figure 2). Some species exhibit strongly determinate shoot extension, in which the terminal meristem is intrinsically downregulated soon after the spring flush and remains inactive for the rest of the season, consistent with a prolonged internal suppression that is hormonally mediated (paradormancy; Lang *et al.*, 1987; Figure 4; e.g. *Fraxinus* spp., *Acer* spp. and *Prunus* spp.). At the other extreme, highly indeterminate species maintain an active shoot apical meristem and can continue producing new leaves and internodes on top of the preformed shoot as long as conditions remain favorable, sometimes stretching their indeterminate growth well into autumn (e.g. *Sequoia* spp., *Cupressus* spp. and *Alnus* spp.). Between these extremes, many species occupy an intermediate position: they continue to produce neoformed leaves and internodes after the preformed increment, but growth cessation is often coordinated by seasonal cues—frequently photoperiod acting through hormonal regulation (e.g. *Pinus* spp., *Betula* spp. and *Populus* spp.; Maurya & Bhalerao, 2017; Triozzi *et al.*, 2018; Böhlenius *et al.*, 2006). Indeterminate growth can occur either through continuous activity of the shoot apical meristem forming new leaves and internodes without buds (free growth), or through the formation and subsequent flushing of buds

mid-season without a period of dormancy (second flushing or lammas growth; Figure 2).

Although this concept is often presented dichotomously (Kozłowski & Pallardy, 1997; Lechowicz, 1984), but see Kikuzawa, 1983; Damascos *et al.*, 2005), with species classified as either determinate or indeterminate growers, most species likely exist along a gradient with numerous intermediate forms. For instance, many temperate oaks (*Quercus spp.*) are considered determinate growers, but can exhibit multiple flushes. Similarly, Douglas-fir (*Pseudotsuga menziesii*) gradually adopts a more determinate growth habit with maturity (Borchert, 1976; Heuret *et al.*, 2006).

Importantly, the degree of determinacy can be quantified at the species level, at least for the shoot apical meristem as the ratio of leaves produced during the relevant growth period to the number of leaf primordia preformed in the bud, with values above 1 indicating increasing neoformation (Box 1). Figure 4 provides a literature-based classification of determinacy for some common species, many of which fall into an intermediate position.

Evolutionary trade-offs: the cost of certainty and flexibility

In most temperate and boreal ecosystems, determinate and indeterminate growth strategies co-occur, indicating that neither strategy is universally superior. Instead, they reflect contrasting solutions to a fundamental trade-off between growth certainty and growth flexibility under seasonal and variable environments.

Indeterminate species are characterized by continued or episodic activity of the shoot apical meristem during the growing season. This does not necessarily imply faster intrinsic growth rates, but rather a greater capacity to extend growth when conditions remain favorable. Such temporal flexibility allows individuals to exploit transient resource availability and recover from partial damage, traits that are advantageous in disturbed or early-successional environments. Accordingly, indeterminate growth is frequently associated with early successional species (Marks, 1975). However, maintaining active meristems later into the season also increases exposure to environmental stressors such as summer drought and early autumn frost, thereby raising the risk of tissue loss or incomplete maturation (Figure 4). As a result, indeterminate growth strategies often involve higher interannual variability in performance and, in some cases, shorter tissue or organismal lifespans (Millet *et al.*, 1999; Brien *et al.*, 2020).

195

196 In contrast, determinate species preform a substantial portion of their shoot growth in advance and
 197 deploy this tissue during a restricted period of the growing season. This strategy limits short-term
 198 plasticity, because the year’s shoot-growth potential is set during bud formation in the preceding year
 199 and is largely realized by expanding preformed organs (Gordon *et al.*, 2006). At the same time, it
 200 reduces exposure of sensitive tissues to late-season stress and synchronizes growth with the most pre-
 201 dictable and favorable part of the season. As a consequence, determinate growth strategies are often
 202 associated with stronger local adaptation and reduced sensitivity to short-term climatic fluctuations
 203 (Leites & Benito Garzón, 2023).

204

205 These contrasting strategies help explain the coexistence of determinate and indeterminate species
 206 within the same landscapes. For example, western larch (*Larix occidentalis*) and Douglas-fir (*Pseudot-*
 207 *suga menziesii*) frequently co-occur despite marked differences in growth determinacy. Western larch
 208 exhibits more flexible, indeterminate shoot growth and performs well in environments shaped by fre-
 209 quent disturbance, whereas Douglas-fir shows more determinate growth and stronger local adaptation,
 210 conferring stability under predictable seasonal regimes (Roskilly & Aitken, 2024). Such differences
 211 suggest niche differentiation along gradients of disturbance frequency, environmental stress, and cli-
 212 matic predictability, rather than outcomes driven by straightforward competitive dominance.

213

214 Together, these patterns indicate that growth determinacy reflects a trade-off between flexibility and
 215 reliability in seasonal environments. Under ongoing climate change—marked by longer growing seasons,
 216 greater climatic variability, and increasing environmental stress in many regions—it remains uncertain
 217 whether determinate or indeterminate growth strategies will be favored. Resolving both how trees
 218 regulate their degree of (in)determinacy and which levels along this trait continuum are advantageous
 219 under future conditions will be critical for predicting shifts in species performance, coexistence, and
 220 forest composition.

221 The performance of (in)determinacy with climate change

222 Spring warming has advanced the onset of leaf emergence by up to a month compared to pre-industrial
 223 times (Vitasse *et al.*, 2022), reflecting an earlier reactivation of shoot apical and cambial meristems
 224 under warmer conditions. In contrast, the cessation of growth in autumn has shifted far less than

225 expected given the extension of climatically favorable conditions (Zani *et al.*, 2020; Zohner *et al.*,
226 2023). As a result, phenological sequences are increasingly observed to shift as a whole toward spring
227 rather than being symmetrically extended at both ends of the growing season (Keenan & Richardson,
228 2015; Meng *et al.*, 2024). This asymmetric response highlights a key constraint relevant to growth
229 (in)determinacy: while growth onset is highly plastic and environmentally responsive, the duration of
230 meristematic activity later in the season appears more tightly regulated by plant internal processes
231 and differs markedly among species. Such constraints on growth cessation help explain why longer
232 growing seasons do not necessarily translate into increased growth or biomass production (Zani *et al.*,
233 2020; Bose *et al.*, 2025; Wolkovich *et al.*, 2025).

234
235 Along the continuum of growth determinacy, we hypothesize that species positioned toward the more
236 determinate end will largely maintain their fixed seasonal growth schedules under climate warming,
237 with limited capacity to extend growth later into the season. In regions where climate change pri-
238 marily increases heat and water stress during mid to late summer, such species are expected to show
239 reduced annual biomass production, because their restricted growth window increasingly overlaps with
240 unfavorable conditions (Figure 3).

241
242 In contrast, we predict that species toward the indeterminate end of the continuum will be more capable
243 of prolonging meristematic activity into late summer and autumn when conditions become favorable
244 again, potentially resulting in higher annual production under some future climate scenarios (Caffarra
245 & Donnelly, 2011; Richardson *et al.*, 2009; Zohner *et al.*, 2020, Figure 3). This capacity for temporal
246 extension of growth is particularly common among early successional species, which tend to leaf out
247 earlier in spring and senesce later in autumn than late-successional species. Their earlier leaf-out is
248 consistent with generally lower chilling and forcing requirements for bud burst and dormancy release
249 (Heide, 1993; Laube *et al.*, 2014; Hu *et al.*, 2022; Baumgarten *et al.*, 2021; Basler & Körner, 2012),
250 and in many cases weaker photoperiod sensitivity (Basler & Körner, 2012).

251
252 Extreme climatic events and increasing environmental stress provide an important context for evaluat-
253 ing these predictions. More determinate growth strategies concentrate growth earlier in the season and
254 may therefore reduce exposure to unfavorable late-season conditions by operating within a relatively
255 narrow spring to early-summer window that often coincides with more favorable temperature and soil

256 moisture conditions (Pearson & D’Orangeville, 2022). In systems where drought or heat stress typi-
257 cally intensifies later in the season, this strategy may provide a wider safety margin against late-season
258 stress, consistent with the ‘late-summer drought avoidance hypothesis’ (Pearson & D’Orangeville, 2022;
259 D’Orangeville *et al.*, 2018).

260 Where climate change introduces novel or intensified drought stress within this window, however, de-
261 terminate species are likely to experience reduced annual production and limited capacity for recovery
262 within the same growing season once canopy development has been compromised (Silvestro *et al.*, 2023,
263 Figure 3).

264
265 Species with more indeterminate growth habits often remain green and physiologically active for longer
266 periods, leafing out early in spring and senescing late in autumn, with deciduous species occasionally
267 retaining functional foliage until the first freezing events (Figure 3). As a consequence, a substantial
268 fraction of their growth period coincides with parts of the growing season during which drought stress
269 is projected to intensify under future climates, particularly in water-limited regions (Figure 3). While
270 this extended activity increases exposure to environmental extremes, it also creates opportunities for
271 compensatory growth when conditions become favorable again.

272
273 In essence, the flexible growth schedule of indeterminately growing species may allow plants to: (1)
274 produce tissue better adapted to harsh environmental conditions as it is formed in the current season
275 (e.g., smaller leaves with higher specific leaf area), and (2) catch up and compensate later in the season
276 by another production boost (Stevens & Lindroth, 2005). Mediterranean ecosystems provide a clear
277 example of this flexibility, where many species suspend growth during summer drought but resume
278 meristematic activity in autumn or winter when conditions allow (Volaire *et al.*, 2023). Evidence that
279 trees can take advantage of such a ‘second growing season’ after summer drought has been found in
280 pines in Mediterranean climates through polycyclic flushing—a form of indeterminate growth (Girard
281 *et al.*, 2011, Figure 3,).

282
283 We therefore hypothesize that climate change will increasingly disrupt established phenological cy-
284 cles, favoring species and genotypes that are more plastic in rearranging their activities by resuming
285 growth and/or restoring reserves (e.g., starch, nutrients) later in the year, thereby recovering from
286 and compensating for stress-induced damage and losses. Such differential capacity to exploit newly

287 available growth opportunities or to recover after stress is expected to intensify competition among
 288 co-occurring species by altering the timing and magnitude of resource uptake, potentially leading to
 289 the reassembly of forest communities toward species occupying more indeterminate positions along the
 290 growth continuum. This shift may be reinforced if climate change increases not only mean stress but
 291 also climatic variability (van der Wiel & Bintanja, 2021), because flexible (plastic) growth schedules
 292 can capitalize on transient favorable windows and compensate after setbacks (Richter *et al.*, 2012;
 293 Chevin *et al.*, 2010; Botero *et al.*, 2015).

294

295 (In)determinacy beyond primary shoot growth

296 Although (in)determinacy is most clearly defined *sensu stricto* at the shoot apical meristem, its rel-
 297 evance may extend beyond primary shoot growth because shoot extension and cambial growth are
 298 coordinated within the plant. Primary growth governs crown expansion and annual leaf deploy-
 299 ment, thereby influencing total leaf area and canopy photosynthetic capacity (Girard *et al.*, 2011;
 300 Soolanayakanahally *et al.*, 2015). Primary shoot growth also establishes the framework within which
 301 later radial growth can occur by determining the extent and geometry of axes along which cambial ac-
 302 tivity is expressed. Empirical studies showing close within-season associations between shoot extension
 303 and stem or radial growth therefore point less to simple control by one process over the other than to
 304 coordinated development between apical and cambial meristems (Cuny *et al.*, 2012; Zhang *et al.*, 2019).

305

306 At the same time, we do not suggest that cambial growth is simply controlled by newly formed leaves
 307 or by current assimilate supply. Primary and secondary growth have partially distinct controls and
 308 constraints, and cause-and-effect relationships between them are not straightforward (Hartmann &
 309 Trumbore, 2016; Trugman & Anderegg, 2025). Rather, from a whole-plant perspective, production of
 310 additional leaves, xylem, and phloem is likely to be coordinated because each is only advantageous if
 311 matched by corresponding hydraulic and carbon-transport demands. In highly indeterminate species,
 312 prolonged primary growth may be part of a broader syndrome in which axis extension and cambial
 313 activity remain active for longer.

314

315 Although vascular development is not preformed in the same mechanistic sense as shoot organs enclosed
 316 in buds, vascular tissues can nonetheless show seasonal carry-over. In particular, where conductive

phloem persists through winter, part of the transport system available early in the next season was established previously and may therefore represent a functional analogue of preformation (Savage, 2020; Savage & Chuine, 2021). The timing and duration of cambial cell division constrain the rate and seasonal extent of secondary growth, while subsequent cell maturation can further restrict wood formation (Lupi *et al.*, 2010). At the same time, cambial cell production can cease even when environmental conditions remain favorable and the canopy is still photosynthetically active, pointing to internal regulation of secondary growth (Buttò *et al.*, 2020; Arend *et al.*, 2024).

In contrast to shoot and cambial growth, below-ground meristems appear to follow a more opportunistic strategy. Root apical meristems can remain active whenever local conditions permit, with rhizotron studies documenting root growth during winter under suitable soil temperatures (Lyford & Wilson, 1966), suggesting that roots often lack a clearly defined dormant phase (Radville *et al.*, 2016; Marchand *et al.*, 2025). However, the extent to which below-ground growth reflects direct environmental control, coordination with above-ground growth, or genetically constrained strategies remains unresolved (Abramoff & Finzi, 2015; Makoto *et al.*, 2020; Silvestro *et al.*, 2025; Campioli *et al.*, 2024).

Together, these observations suggest that although (in)determinacy is most rigorously defined at the shoot apical meristem, shoot-level variation may still provide a useful lens for studying coordination among plant growth processes and their consequences for whole-plant performance.

Future directions

A central challenge emerging from this perspective is to better understand how variation along the continuum of growth (in)determinacy shapes tree performance under changing climatic conditions. Rather than representing discrete strategies, degrees of determinacy define bounded trait spaces that differ among species but can vary with age, provenance, and environmental context. While the degree of determinacy is largely constrained at the species level, evidence from provenance trials shows meaningful intraspecific variation; for example, populations from lower latitudes often exhibit higher frequencies of second flushes than those from northern origins (Rudolph, 1964). Clarifying how these species-specific bounds and population-level adjustments interact with warming, drought, and increasing climatic variability will be critical for understanding how forest communities respond and reassemble under climate

change. Extending this framework beyond woody perennials to include herbaceous species and perennial grasses may further help test the generality of (in)determinacy as a developmental constraint on growth timing, while highlighting differences arising from life form and the absence of secondary growth.

One key knowledge gap concerns how different degrees of (in)determinacy help trees avoid or buffer environmental stress while still maintaining the capacity to resume growth, repair tissues, replenish reserves, and compensate for earlier losses within the same season. Resolving the trade-off between escaping unfavorable conditions and exploiting longer or intermittently favorable growing periods is essential for anticipating how forest communities will change over the coming decades.

Understanding whether shoot-level (in)determinacy is associated with analogous patterns of developmental carry-over or seasonal flexibility in other organs may provide insight into the evolution of this trait, especially through cross-species analyses. Both growth forms seem to occur across most species of the same family or clade and hence appear to have evolved repeatedly in different groups (Figure 4), making it difficult to speculate on which strategy is ancestral (but see Hariharan *et al.*, 2016) and how rapidly (in)determinacy can evolve. As data on the degree of (in)determinacy across species accumulates (see Box 1), such analyses could potentially provide insights into this and also aid forecasting for unsampled species.

Finally, future work should aim to link the temporal dynamics of primary (shoot and root) and secondary (cambial) meristems with patterns of reserve storage and reproductive allocation. Integrating shoot increment formation, wood production, non-structural carbohydrate dynamics, and reproductive effort within a common temporal framework will be essential for understanding whole-tree carbon allocation. Correlating annual tree-ring formation with shoot increments—while accounting for the timing of inter-annual phytomer production and distinguishing between preformed and neoformed tissues (Figure 2)—offers a promising path toward such integration and toward improving mechanistic predictions of tree growth and forest carbon dynamics under future climates.

Acknowledgments

This text emerged from a grant of the Swiss National Foundation to F.B. (grant number P500PB_210943) and was supported by CFI and NSERC Discovery funding to EMW. We thank J. Ngo for designing

377 figure 2.

Bibliography

- Abramoff, R.Z. & Finzi, A.C. (2015). Are above- and below-ground phenology in sync? *New Phytologist*, 205, 1054–1061.
- Alvarado, M.V., Terrazas, T. & Rojas-Leal, A. (2025). Cambial activity in five species of the tropical dry forest: Dynamics of seasonal changes in the cambial region. *Trees*, 39, 51.
- Arend, M., Hoch, G. & Kahmen, A. (2024). Stem growth phenology, not canopy greening, constrains deciduous tree growth. *Tree Physiology*, 44, tpad160.
- Barthélémy, D. & Caraglio, Y. (2007). Plant Architecture: A Dynamic, Multilevel and Comprehensive Approach to Plant Form, Structure and Ontogeny. *Annals of Botany*, 99, 375–407.
- Basler, D. & Körner, C. (2012). Photoperiod sensitivity of bud burst in 14 temperate forest tree species. *Agricultural and Forest Meteorology*, 165, 73–81.
- Baumgarten, F., Gessler, A. & Vitasse, Y. (2023). No risk—no fun: Penalty and recovery from spring frost damage in deciduous temperate trees. *Functional Ecology*, 37, 648–663.
- Baumgarten, F., Zohner, C.M., Gessler, A. & Vitasse, Y. (2021). Chilled to be forced: The best dose to wake up buds from winter dormancy. *New Phytologist*, 230, 1366–1377.
- Böhlenius, H., Huang, T., Charbonnel-Campaa, L., Brunner, A.M., Jansson, S., Strauss, S.H. & Nilsson, O. (2006). CO/FT Regulatory Module Controls Timing of Flowering and Seasonal Growth Cessation in Trees. *Science*, 312, 1040–1043.
- Borchert, R. (1976). The concept of juvenility in woody plants. *Acta Horticulturae*, pp. 21–36.
- Borchert, R. (1994). Soil and Stem Water Storage Determine Phenology and Distribution of Tropical Dry Forest Trees. *Ecology*, 75, 1437–1449.
- Bose, A.K., Etzold, S., Meusburger, K., Gessler, A., Baltensweiler, A., Braun, S., Buchmann, N., Camarero, J.J., Haeni, M., Kahmen, A., Peters, R.L., Sterck, F.J., Tresch, S., Walthert, L. & Zweifel, R. (2025). Decreasing Stem Growth in Common European Tree Species Despite Earlier Growth Onset. *Global Change Biology*, 31, e70318.

403 Botero, C.A., Weissing, F.J., Wright, J. & Rubenstein, D.R. (2015). Evolutionary tipping points in
 404 the capacity to adapt to environmental change. *Proceedings of the National Academy of Sciences*,
 405 112, 184–189.

406 Brienens, R.J.W., Caldwell, L., Duchesne, L., Voelker, S., Barichivich, J., Baliva, M., Ceccantini, G.,
 407 Di Filippo, A., Helama, S., Locosselli, G.M., Lopez, L., Piovesan, G., Schöngart, J., Villalba, R.
 408 & Gloor, E. (2020). Forest carbon sink neutralized by pervasive growth-lifespan trade-offs. *Nature*
 409 *Communications*, 11, 4241.

410 Buttò, V., Shishov, V., Tychkov, I., Popkova, M., He, M., Rossi, S., Deslauriers, A. & Morin, H. (2020).
 411 Comparing the Cell Dynamics of Tree-Ring Formation Observed in Microcores and as Predicted by
 412 the Vaganov–Shashkin Model. *Frontiers in Plant Science*, 11.

413 Caffarra, A. & Donnelly, A. (2011). The ecological significance of phenology in four different tree
 414 species: Effects of light and temperature on bud burst. *International Journal of Biometeorology*, 55,
 415 711–721.

416 Campioli, M., Marchand, L.J., Zahnd, C., Zuccarini, P., McCormack, M.L., Landuyt, D., Lorer, E.,
 417 Delpierre, N., Gričar, J. & Vitasse, Y. (2024). Environmental Sensitivity and Impact of Climate
 418 Change on leaf-, wood- and root Phenology for the Overstory and Understory of Temperate Decid-
 419 uous Forests. *Current Forestry Reports*, 11, 1.

420 Chapin, F.S., Schulze, E. & Mooney, H.A. (1990). The Ecology and Economics of Storage in Plants.
 421 *Annual Review of Ecology and Systematics*, 21, 423–447.

422 Chevin, L.M., Lande, R. & Mace, G.M. (2010). Adaptation, Plasticity, and Extinction in a Changing
 423 Environment: Towards a Predictive Theory. *PLOS Biology*, 8, e1000357.

424 Cuny, H.E., Rathgeber, C.B., Lebourgeois, F., Fortin, M. & Fournier, M. (2012). Life strategies in
 425 intra-annual dynamics of wood formation: Example of three conifer species in a temperate forest in
 426 north-east France. *Tree Physiology*, 32, 612–625.

427 Damascos, M.A., Prado, C.H.B.A. & Ronquim, C.C. (2005). Bud Composition, Branching Patterns
 428 and Leaf Phenology in Cerrado Woody Species. *Annals of Botany*, 96, 1075–1084.

429 Delpierre, N., Vitasse, Y., Chuine, I., Guillemot, J., Bazot, S., Rutishauser, T. & Rathgeber, C.B.K.
 430 (2016). Temperate and boreal forest tree phenology: From organ-scale processes to terrestrial ecosys-
 431 tem models. *Annals of Forest Science*, 73, 5–25.

432 D'Orangeville, L., Maxwell, J., Kneeshaw, D., Pederson, N., Duchesne, L., Logan, T., Houle, D.,
433 Arseneault, D., Beier, C.M., Bishop, D.A., Druckenbrod, D., Fraver, S., Girard, F., Halman, J.,
434 Hansen, C., Hart, J.L., Hartmann, H., Kaye, M., Leblanc, D., Manzoni, S., Ouimet, R., Rayback, S.,
435 Rollinson, C.R. & Phillips, R.P. (2018). Drought timing and local climate determine the sensitivity
436 of eastern temperate forests to drought. *Global Change Biology*, pp. n/a–n/a.

437 Ejsmond, M.J., Czarnoński, M., Kapustka, F. & Kozłowski, J. (2010). How to Time Growth and
438 Reproduction during the Vegetative Season: An Evolutionary Choice for Indeterminate Growers in
439 Seasonal Environments. *The American Naturalist*, 175, 551–563.

440 Estiarte, M. & Peñuelas, J. (2015). Alteration of the phenology of leaf senescence and fall in winter
441 deciduous species by climate change: Effects on nutrient proficiency. *Global Change Biology*, 21,
442 1005–1017.

443 Etzold, S., Sterck, F., Bose, A.K., Braun, S., Buchmann, N., Eugster, W., Gessler, A., Kahmen, A.,
444 Peters, R.L. & Vitasse, Y. (2021). Number of growth days and not length of the growth period
445 determines radial stem growth of temperate trees. *Ecology Letters*.

446 Girard, F., Vennetier, M., Ouarmim, S., Caraglio, Y. & Misson, L. (2011). Polycyclism, a fundamental
447 tree growth process, decline with recent climate change: The example of *Pinus halepensis* Mill. in
448 Mediterranean France. *Trees*, 25, 311–322.

449 Gordon, D., Damiano, C. & DeJong, T.M. (2006). Preformation in vegetative buds of *Prunus persica*:
450 Factors influencing number of leaf primordia in overwintering buds. *Tree Physiology*, 26, 537–544.

451 Guédon, Y., Puntieri, J.G., Sabatier, S. & Barthélémy, D. (2006). Relative Extents of Preformation
452 and Neoformation in Tree Shoots: Analysis by a Deconvolution Method. *Annals of Botany*, 98,
453 835–844.

454 Hallé, F., Oldeman, R.A.A. & Tomlinson, P.B. (1978). *Tropical Trees and Forests*. Springer, Berlin,
455 Heidelberg.

456 Hariharan, I.K., Wake, D.B. & Wake, M.H. (2016). Indeterminate growth: Could it represent the
457 ancestral condition? *Cold Spring Harbor Perspectives in Biology*, 8, a019174.

458 Hartmann, H. & Trumbore, S. (2016). Understanding the roles of nonstructural carbohydrates in
459 forest trees – from what we can measure to what we want to know. *New Phytologist*, 211, 386–403.

460 Heide, O.M. (1993). Daylength and thermal time responses of budburst during dormancy release in
 461 some northern deciduous trees. *Physiologia Plantarum*, 88, 531–540.

462 Heuret, P., Meredieu, C., Coudurier, T., Courdier, F. & Barthélémy, D. (2006). Ontogenetic trends
 463 in the morphological features of main stem annual shoots of *Pinus pinaster* (Pinaceae). *American*
 464 *Journal of Botany*, 93, 1577–1587.

465 Hover, A., Buissart, F., Caraglio, Y., Heinz, C., Pailler, F., Ramel, M., Vennetier, M., Prévosto, B.
 466 & Sabatier, S. (2017). Growth phenology in *Pinus halepensis* Mill.: Apical shoot bud content and
 467 shoot elongation. *Annals of Forest Science*, 74, 39.

468 Hsiao, T.C. (1973). Plant responses to water stress. *Annual review of plant physiology*, 24, 519–570.

469 Hu, Z., Wang, H., Dai, J., Ge, Q. & Lin, S. (2022). Stronger Spring Phenological Advance in Future
 470 Warming Scenarios for Temperate Species With a Lower Chilling Sensitivity. *Frontiers in Plant*
 471 *Science*, 13.

472 Huijser, P. & Schmid, M. (2011). The control of developmental phase transitions in plants. *Develop-*
 473 *ment*, 138, 4117–4129.

474 Jin, S., Zhang, W., Shao, J., Wan, P., Cheng, S., Cai, S., Yan, G. & Li, A. (2022). Estimation of Larch
 475 Growth at the Stem, Crown, and Branch Levels Using Ground-Based LiDAR Point Cloud. *Journal*
 476 *of Remote Sensing*, 2022.

477 Karkach, A.S. (2006). Trajectories and models of individual growth. *Demographic Research*, 15, 347–
 478 400.

479 Keenan, T.F. & Richardson, A.D. (2015). The timing of autumn senescence is affected by the timing
 480 of spring phenology: Implications for predictive models. *Global Change Biology*, 21, 2634–2641.

481 Kikuzawa, K. (1983). Leaf survival of woody plants in deciduous broad-leaved forests. 1. Tall trees.
 482 *Canadian Journal of Botany*, 61, 2133–2139.

483 Körner, C. (2008). Winter crop growth at low temperature may hold the answer for alpine treeline
 484 formation. *Plant Ecology & Diversity*, 1, 3–11.

485 Kovaleski, A.P., Londo, J.P. & Finkelstein, K.D. (2019). X-ray phase contrast imaging of *Vitis* spp.
 486 buds shows freezing pattern and correlation between volume and cold hardiness. *Scientific Reports*,
 487 9, 14949.

488 Kozłowski, T.T. (2012). *Seed Germination, Ontogeny, and Shoot Growth*. Elsevier.

489 Kozłowski, T.T. & Pallardy, S.G. (1997). *Growth Control in Woody Plants*. Elsevier.

490 Kuster, T.M., Dobbertin, M., Günthardt-Goerg, M.S., Schaub, M. & Arend, M. (2014). A Phenological
491 Timetable of Oak Growth under Experimental Drought and Air Warming. *PLoS ONE*, 9, e89724.

492 Lang, G.A., Early, J.D., Martin, G.C. & Darnell, R.L. (1987). Endo-, Para-, and Ecodormancy:
493 Physiological Terminology and Classification for Dormancy Research. *HortScience*, 22, 371–377.

494 Laube, J., Sparks, T.H., Estrella, N., Hoffer, J., Ankerst, D.P. & Menzel, A. (2014). Chilling outweighs
495 photoperiod in preventing precocious spring development. *Global Change Biology*, 20, 170–82.

496 Lechowicz, M.J. (1984). Why Do Temperate Deciduous Trees Leaf Out at Different Times? Adaptation
497 and Ecology of Forest Communities. *The American Naturalist*.

498 Leites, L. & Benito Garzón, M. (2023). Forest tree species adaptation to climate across biomes:
499 Building on the legacy of ecological genetics to anticipate responses to climate change. *Global*
500 *Change Biology*, 29, 4711–4730.

501 Luo, T., Liu, X., Zhang, L., Li, X., Pan, Y. & Wright, I.J. (2018). Summer solstice marks a seasonal
502 shift in temperature sensitivity of stem growth and nitrogen-use efficiency in cold-limited forests.
503 *Agricultural and Forest Meteorology*, 248, 469–478.

504 Lupi, C., Morin, H., Deslauriers, A. & Rossi, S. (2010). Xylem phenology and wood production:
505 Resolving the chicken-or-egg dilemma. *Plant, Cell & Environment*, 33, 1721–1730.

506 Lyford, W. & Wilson, B. (1966). Controlled growth of forest tree roots: Technique and application.
507 *Harvard Forest Paper*.

508 Makoto, K., Wilson, S.D., Sato, T., Blume-Werry, G. & Cornelissen, J.H.C. (2020). Synchronous and
509 asynchronous root and shoot phenology in temperate woody seedlings. *Oikos*, 129, 643–650.

510 Marchand, L.J., Gričar, J., Zuccarini, P., Dox, I., Mariën, B., Verlinden, M., Heinecke, T., Prislan,
511 P., Marie, G. & Lange, H. (2025). No winter halt in below-ground wood growth of four angiosperm
512 deciduous tree species. *Nature Ecology & Evolution*, pp. 1–9.

513 Marks, P.L. (1975). On the Relation between Extension Growth and Successional Status of Deciduous
514 Trees of the Northeastern United States. *Bulletin of the Torrey Botanical Club*, 102, 172–177.

Maurya, J.P. & Bhalerao, R.P. (2017). Photoperiod- and temperature-mediated control of growth
cessation and dormancy in trees: A molecular perspective. *Annals of Botany*, 120, 351–360.

McDaniel, C.N. (1992). Induction and Determination: Developmental Concepts. *Flowering Newsletter*,
pp. 3–6.

Meng, F., Felton, A.J., Mao, J., Cong, N., Smith, W.K., Körner, C., Hu, Z., Hong, S., Knott, J., Yan,
Y., Guo, B., Deng, Y., Leisz, S., Dorji, T., Wang, S. & Chen, A. (2024). Consistent time allocation
fraction to vegetation green-up versus senescence across northern ecosystems despite recent climate
change. *Science Advances*, 10, eadn2487.

Millet, J., Bouchard, A. & Édelin, C. (1999). Relationship between architecture and successional status
of trees in the temperate deciduous forest. *Écoscience*, 6, 187–203.

Moore, E. (1909). The Study of Winter Buds with Reference to Their Growth and Leaf Content.
Bulletin of the Torrey Botanical Club, 36, 117–145.

O’Sullivan, O.S., Heskell, M.A., Reich, P.B., Tjoelker, M.G., Weerasinghe, L.K., Penillard, A., Zhu,
L., Egerton, J.J.G., Bloomfield, K.J., Creek, D., Bahar, N.H.A., Griffin, K.L., Hurry, V., Meir, P.,
Turnbull, M.H. & Atkin, O.K. (2017). Thermal limits of leaf metabolism across biomes. *Global
Change Biology*, 23, 209–223.

Pearson, E. & D’Orangeville, L. (2022). Relating the Growth Phenology and Biomass Allocation in
Seedlings of 13 Acadian Tree Species With Their Drought Tolerance. *Frontiers in Forests and Global
Change*, 5.

Poethig, R.S. (2003). Phase Change and the Regulation of Developmental Timing in Plants. *Science*,
301, 334–336.

Pugnaire, F.I., Serrano, LUIS. & Pardos, JOSE. (1999). Constraints by water stress on plant growth.
Handbook of plant and crop stress, 2, 271–283.

Radville, L., McCormack, M.L., Post, E. & Eissenstat, D.M. (2016). Root phenology in a changing
climate. *Journal of Experimental Botany*, 67, 3617–3628.

Ramos, D.M., Andrade, J.M., Alberton, B.C., Moura, M.S.B., Domingues, T.F., Neves, N., Lima,
J.R.S., Souza, R., Souza, E., Silva, J.R., Espírito-Santo, M.M., Morellato, L.P.C. & Cunha, J.

- (2023). Multiscale phenology of seasonally dry tropical forests in an aridity gradient. *Frontiers in Environmental Science*, 11.
- Richardson, A.D., Hollinger, D.Y., Dail, D.B., Lee, J.T., Munger, J.W. & O’keefe, J. (2009). Influence of spring phenology on seasonal and annual carbon balance in two contrasting New England forests. *Tree Physiology*, 29, 321–331.
- Richter, S., Kipfer, T., Wohlgemuth, T., Calderón Guerrero, C., Ghazoul, J. & Moser, B. (2012). Phenotypic plasticity facilitates resistance to climate change in a highly variable environment. *Oecologia*, 169, 269–279.
- Roskilly, B. & Aitken, S. (2024). Weak Local Adaptation to Climate in Seedlings of a Deciduous Conifer Suggests Limited Benefits and Risks of Assisted Gene Flow. *Evolutionary Applications*, 17, e70001.
- Rossi, S., Deslauriers, A., Anfodillo, T., Morin, H., Saracino, A., Motta, R. & Borghetti, M. (2006). Conifers in cold environments synchronize maximum growth rate of tree-ring formation with day length. *New Phytologist*, 170, 301–310.
- Rossi, S., Deslauriers, A., Gricar, J., Seo, J.W., Rathgeber, C.B.K., Anfodillo, T., Morin, H., Levanic, T., Oven, P. & Jalkanen, R. (2008). Critical temperatures for xylogenesis in conifers of cold climates. *Global Ecology and Biogeography*, 17, 696–707.
- Rudolph, T.D. (1964). Lammas Growth and Prolepsis in Jack Pine in the Lake States. *Forest Science*, 10, a0001–70.
- Sakai, A. & Larcher, W. (1987). *Freezing Injuries in Plants*, Springer Berlin Heidelberg, Berlin, Heidelberg, vol. 62, pp. 39–58.
- Savage, J.A. (2020). It’s all about timing—or is it? Exploring the potential connection between phloem physiology and whole plant phenology. *American Journal of Botany*, 107, 848–851.
- Savage, J.A. & Chuine, I. (2021). Coordination of spring vascular and organ phenology in deciduous angiosperms growing in seasonally cold climates. *New Phytologist*, 230, 1700–1715.
- Schenker, G., Lenz, A., Körner, C. & Hoch, G. (2014). Physiological minimum temperatures for root growth in seven common European broad-leaved tree species. *Tree Physiology*, 34, 302–313.

569 Silvestro, R., Deslauriers, A., Prislan, P., Rademacher, T., Rezaie, N., Richardson, A.D., Vitasse, Y.
570 & Rossi, S. (2025). From Roots to Leaves: Tree Growth Phenology in Forest Ecosystems. *Current*
571 *Forestry Reports*, 11, 12.

572 Silvestro, R., Zeng, Q., Buttò, V., Sylvain, J.D., Drolet, G., Mencuccini, M., Thiffault, N., Yuan, S.
573 & Rossi, S. (2023). A longer wood growing season does not lead to higher carbon sequestration.
574 *Scientific Reports*, 13, 4059.

575 Smith, S.A. & Brown, J.W. (2018). Constructing a broadly inclusive seed plant phylogeny. *American*
576 *Journal of Botany*, 105, 302–314.

577 Soolanayakanahally, R.Y., Guy, R.D., Street, N.R., Robinson, K.M., Silim, S.N., Albrechtsen, B.R.
578 & Jansson, S. (2015). Comparative physiology of allopatric *Populus* species: Geographic clines in
579 photosynthesis, height growth, and carbon isotope discrimination in common gardens. *Frontiers in*
580 *plant science*, 6, 528.

581 Stearns, S.C. (1989). Trade-Offs in Life-History Evolution. *Functional Ecology*, 3, 259.

582 Stearns, S.C. (1998). *The Evolution Of Life Histories*. Oxford University Press.

583 Stevens, M.T. & Lindroth, R.L. (2005). Induced resistance in the indeterminate growth of aspen
584 (*Populus tremuloides*). *Oecologia*, 145, 297–305.

585 Triozzi, P.M., Ramos-Sánchez, J.M., Hernández-Verdeja, T., Moreno-Cortés, A., Allona, I. & Perales,
586 M. (2018). Photoperiodic Regulation of Shoot Apical Growth in Poplar. *Frontiers in Plant Science*,
587 9, 1030.

588 Trugman, A.T. & Anderegg, L.D.L. (2025). Source vs sink limitations on tree growth: From phys-
589 iological mechanisms to evolutionary constraints and terrestrial carbon cycle implications. *New*
590 *Phytologist*, 245, 966–981.

591 van der Wiel, K. & Bintanja, R. (2021). Contribution of climatic changes in mean and variability to
592 monthly temperature and precipitation extremes. *Communications Earth & Environment*, 2, 1.

593 Vitasse, Y., Baumgarten, F., Zohner, C.M., Rutishauser, T., Pietragalla, B., Gehrig, R., Dai, J., Wang,
594 H., Aono, Y. & Sparks, T.H. (2022). The great acceleration of plant phenological shifts. *Nature*
595 *Climate Change*, 12, 300–302.

596 Volaire, F., Barkaoui, K., Grémillet, D., Charrier, G., Dangles, O., Lamarque, L.J., Martin-StPaul,
597 N. & Chuine, I. (2023). Is a seasonally reduced growth potential a convergent strategy to survive
598 drought and frost in plants? *Annals of Botany*, 131, 245–254.

599 Wang, Q., Liu, W., Leung, C.C., Tarté, D.A. & Gendron, J.M. (2024). Plants distinguish different
600 photoperiods to independently control seasonal flowering and growth. *Science*, 383, eadg9196.

601 Wolkovich, E.M., Ettinger, A.K., Chin, A.R., Chamberlain, C.J., Baumgarten, F., Pradhan, K., Man-
602 zanedo, R.D. & Hille Ris Lambers, J. (2025). Why longer seasons with climate change may not
603 increase tree growth. *Nature Climate Change*, 15, 1283–1292.

604 Zani, D., Crowther, T.W., Mo, L., Renner, S.S. & Zohner, C.M. (2020). Increased growing-season
605 productivity drives earlier autumn leaf senescence in temperate trees. *Science*, 370, 1066–1071.

606 Zhang, Y., Wu, H. & Yang, W. (2019). Forests Growth Monitoring Based on Tree Canopy 3D Recon-
607 struction Using UAV Aerial Photogrammetry. *Forests*, 10, 1052.

608 Zohner, C.M., Mirzagholi, L., Renner, S.S., Mo, L., Rebindaine, D., Bucher, R., Palouš, D., Vitasse,
609 Y., Fu, Y.H., Stocker, B.D. & Crowther, T.W. (2023). Effect of climate warming on the timing of
610 autumn leaf senescence reverses after the summer solstice. *Science*, 381, eadf5098.

611 Zohner, C.M., Mo, L., Pugh, T.A.M., Bastin, J.F. & Crowther, T.W. (2020). Interactive climate
612 factors restrict future increases in spring productivity of temperate and boreal trees. *Global Change*
613 *Biology*, 26, 4042–4055.

614 Zohner, C.M., Renner, S.S., Sebald, V. & Crowther, T.W. (2021). How changes in spring and autumn
615 phenology translate into growth-experimental evidence of asymmetric effects. *Journal of Ecology*,
616 109, 2717–2728.

Box 1: Metrics and proxies related to (in)determinacy

To advance our understanding of growth strategies under climate change, we need not only improved methods for quantifying the degree of (in)determinacy—but also empirical approaches that move beyond a simple binary classification. An organ-level metric at the shoot apical meristem captures the defining features of (in)determinacy and allows it to be explicitly defined and quantified. Temporal dynamics of shoot growth provide a related but indirect proxy that can be applied from individual shoots to crowns and entire forest stands.

Ratio of preformed to neoformed organs: The ratio of the number of leaves (or leaf scars) produced during a growing season to the number of leaf primordia preformed in dormant buds provides a direct, mechanistic, and quantitative measure of shoot-level (in)determinacy. Values close to one indicate growth constrained to preformed organs, whereas values greater than one reflect the production of neoformed organs and thus indeterminate development. This ratio should be interpreted relative to the relevant growth period considered. In species with more than one distinct growth phase within a year, estimates may need to integrate all flushes contributing to annual shoot development, or be reported separately for distinct flushes. Because the expression of neoformation depends on environmental conditions, indeterminate species may appear more determinate in short or unfavourable years. Preformation and neoformation have been documented for more than a century (Moore, 1909), and several studies have quantified the number of preformed primordia and compared it with the final number of emerged leaves or phytomers (Damascos *et al.*, 2005; Kikuzawa, 1983; Guédon *et al.*, 2006). Quantification typically relies on bud dissection, but emerging non-destructive approaches such as micro-CT imaging offer promising alternatives for assessing bud structure and preformation (Kovaleski *et al.*, 2019). This metric remains the most conceptually robust approach for defining and quantifying the level of (in)determinacy across species.

Seasonal growth dynamics of the shoot apical meristem: Temporal patterns of shoot elongation provide a complementary perspective on (in)determinacy at the shoot apical meristem. Across species, elongation may occur over a brief period, extend continuously through the growing season, or appear in multiple distinct growth phases, reflecting differences in how apical meristem activity is maintained or reactivated. Such dynamics can be quantified using repeated shoot measurements or phenological observations, and increasingly through high-resolution terrestrial LiDAR and photogrammetric techniques that capture shoot elongation and structural change at the scale of entire crowns or stands (Zhang *et al.*, 2019; Jin *et al.*, 2022). While temporal growth dynamics do not distinguish preformed from neoformed organs directly, they remain closely linked to shoot apical meristem activity and provide a scalable proxy for the seasonal expression of (in)determinacy. Repeated measurements also allow shoot-elongation curves to be reconstructed, whose shape differs systematically among species (single, short sigmoid vs. prolonged or multi-phase extension; Hover *et al.* (2017); Kuster *et al.* (2014)). Specifically, curve-derived parameters such as the duration of shoot expansion relative to the growing season provide a quantitative proxy for the degree of (in)determinacy.

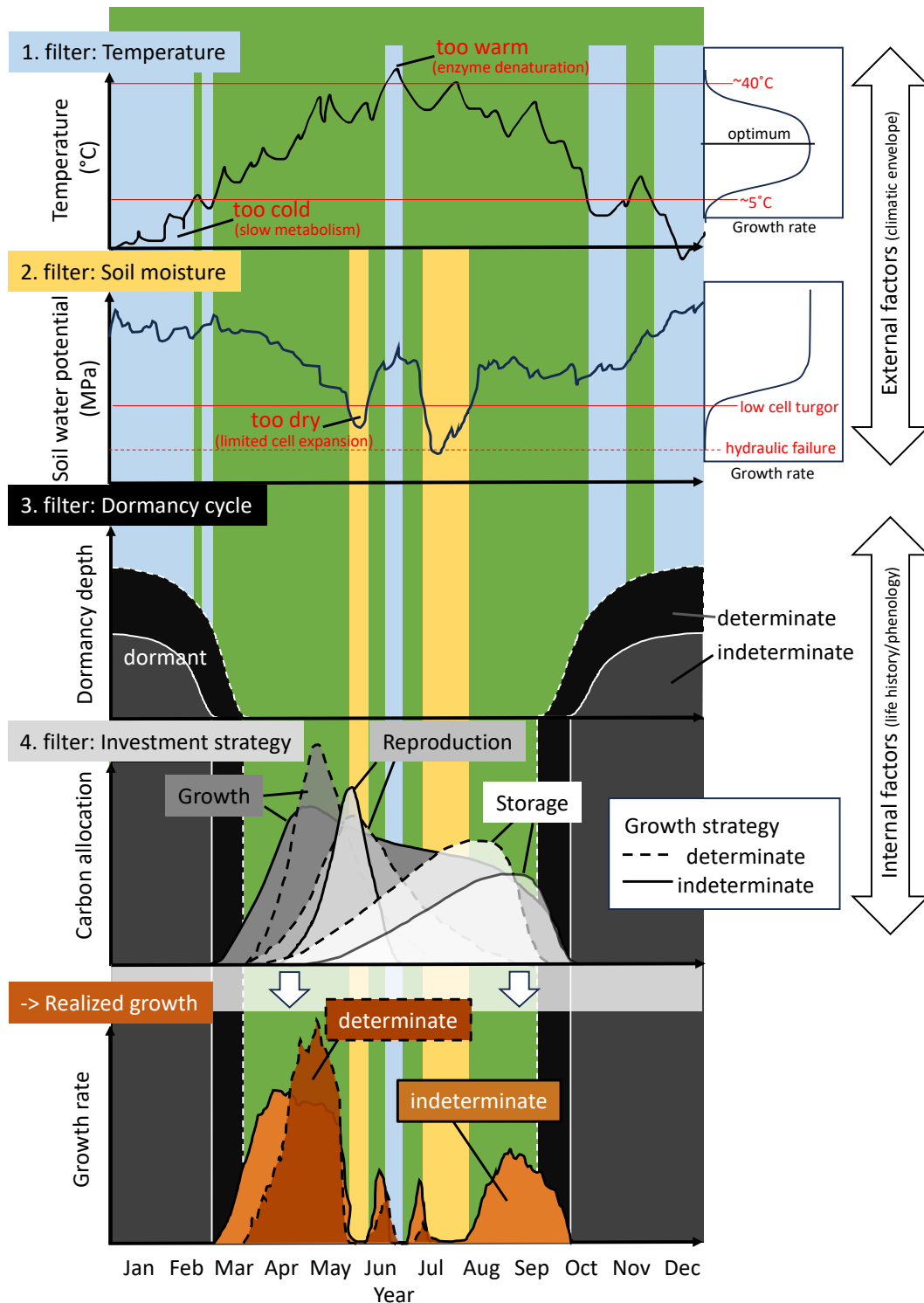


Figure 1: Schematic overview of the discrepancy between the potential growing season and the effectively realized period of vegetative growth (shoot and possibly cambial growth). Green indicates the potential growing season, whereas overlaid coloured areas represent portions of this period that are constrained by different external and internal filters. Environmental factors, such as temperature and soil moisture, can narrow the window of opportunity for vegetative growth when conditions fall outside growth-promoting thresholds (potential light and photoperiodic constraints are not shown). Species-specific phenology can impose an additional filter through dormancy and through prioritization of other developmental processes (e.g. reproduction and storage). The degree of (in)determinacy is illustrated here as two extremes, although we propose that this trait is better understood as continuous.

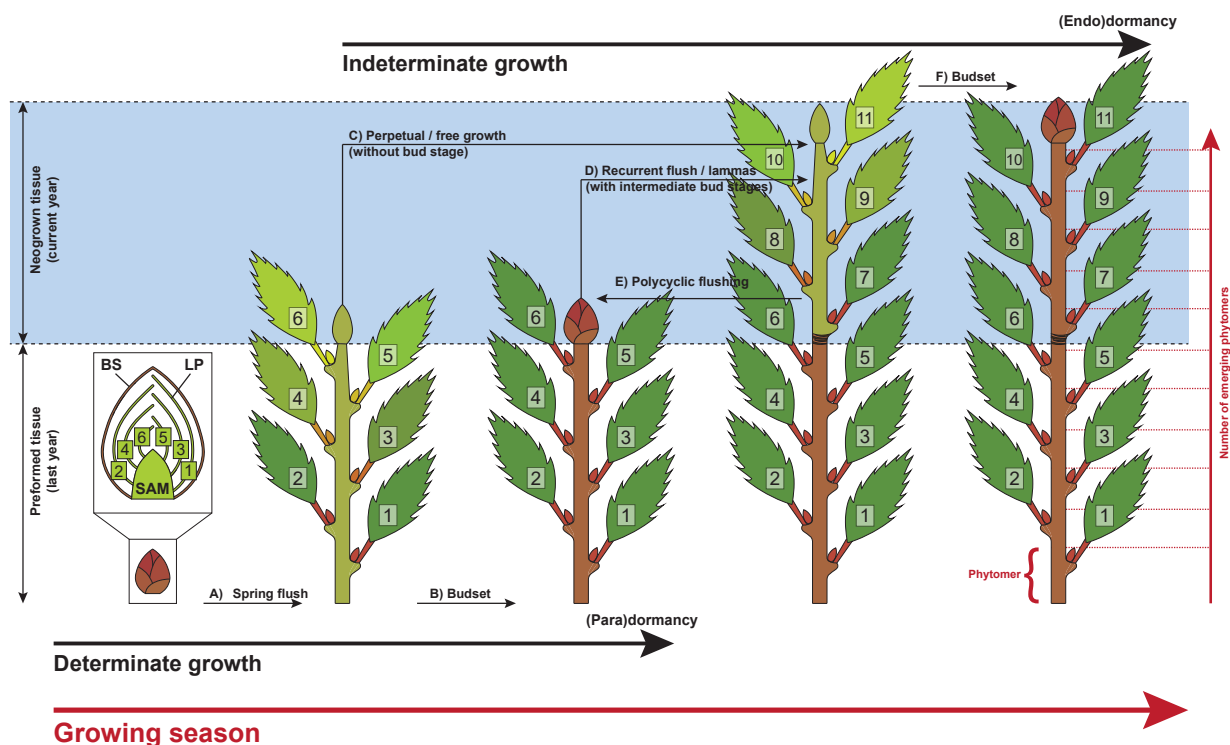


Figure 2: Determinate and indeterminate growth within one growing season for species producing terminal buds. Commonly all tree species deploy buds during their first (spring) flush from prebuilt and overwintering leaf primordia (LP) protected by bud scales (BS; A). Determinate growing species set buds (B) that are under hormonal suppression to inhibit any further activity of the shoot apical meristem – SAM (paradormancy). The basic unit of a shoot is the phytomer which is composed of a node, a leaf, the axillary bud and an internode (see bottom right). Indeterminate growing species continue to produce new phytomers directly (C) or through one (D) to several (E) intermediate bud stage(s). Finally, most species set their terminal buds (F) and enter full dormancy (endodormancy).

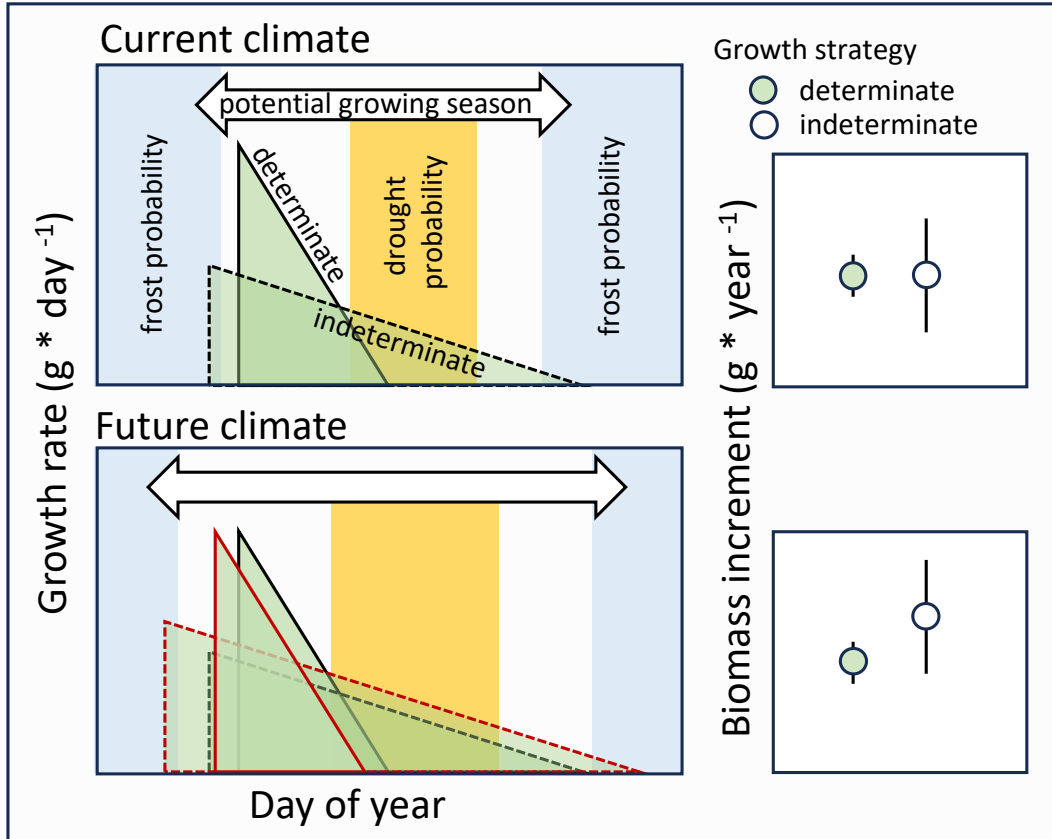


Figure 3: Hypothesized predictions of growth rates under current (upper panels) and future (lower panels) climate for determinate and indeterminate growing species. Note that the indeterminate strategy is more exposed to the risk of frost and drought events while the determinate strategy condenses most growth within a safe period. In the current climate the indeterminate strategy is in balance with benefiting from the full climatic growing season in some years with some drawbacks in other years, resulting in the same mean yearly biomass increment, but with a higher variation (right panel). In a future climate the indeterminate strategy might benefit from longer growing seasons, resulting in an overall higher mean annual biomass increment compared to determinate growers, but this will be dependent on the severity of drought that overlaps the potential growth period of determinate species. Triangle outlines represent the activity window for a current (black) versus a future (red) climate.

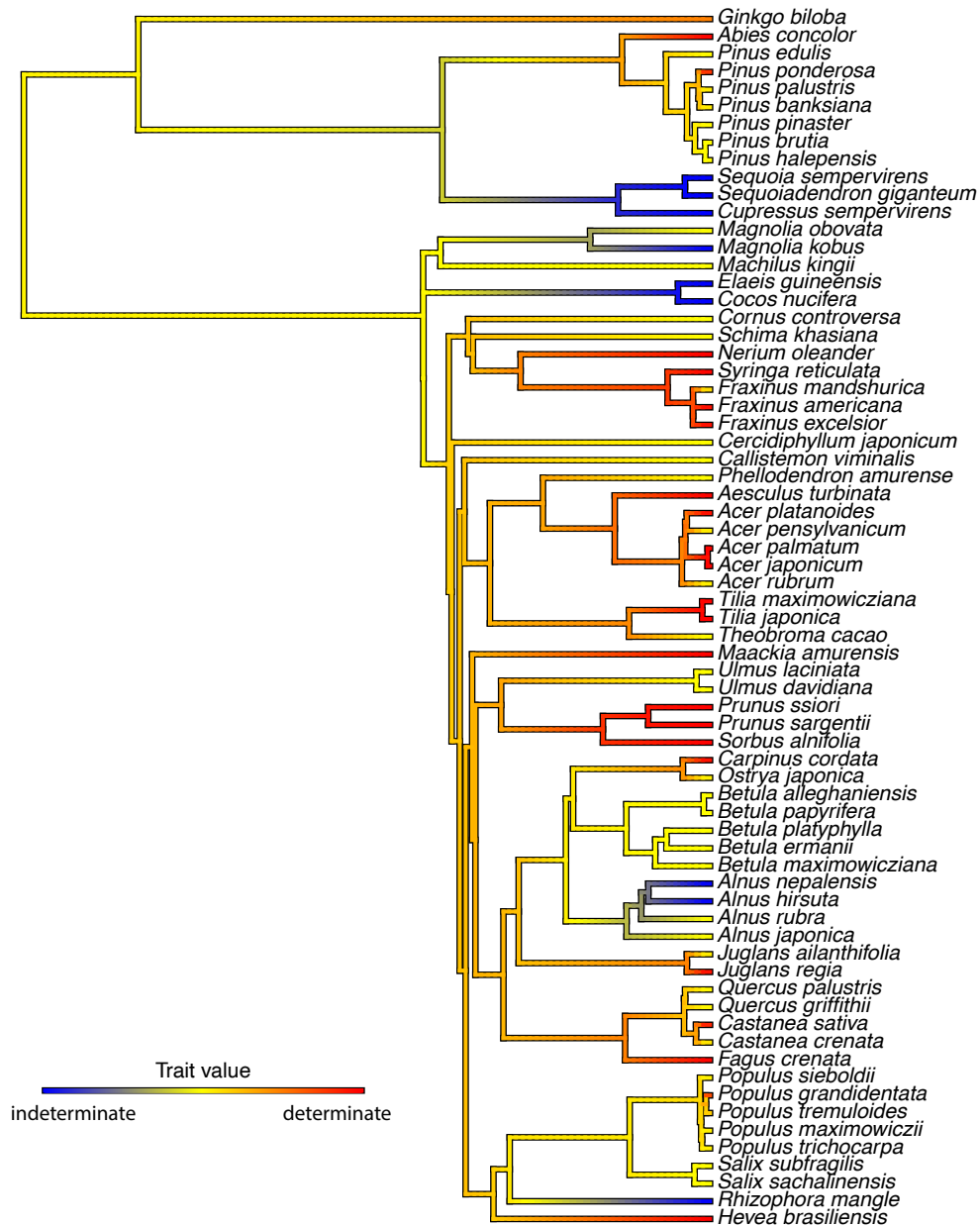


Figure 4: Phylogenetic tree of plant species colored by growth determinacy trait values, visually representing the evolutionary distribution of growth strategies across taxa. This tree includes species for which shoot growth determinacy data were compiled (mainly from Hallé *et al.* (1978) and Kikuzawa (1983)), with the phylogeny pruned from the comprehensive seed plant tree of Smith & Brown (2018) to match the trait dataset. Growth determinacy was scored as indeterminate (blue), intermediate (yellow) and determinate (red), with mean values calculated per species when multiple entries existed. These values were then mapped as a continuous trait onto the phylogeny using the `contMap()` function from the `phytools` R package.